DSP in VLSI

Final Project

Eigen-value Decomposition by Iterative QR Operations

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Name

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Development Environment

HDL	SystemVerilog
Simulation Tools	Synopsys VCS and Verdi
Synthesis Tool	Synopsys DC Compiler
Process	TSMC 16nm (ADFP)
Verification	Python

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Contents

Section 1: Wordlength and RMSE Analysis

1.1 Minimum Fractional Wordlength & CORDIC Stage Selection

To set the fixed-point parameters of the EVD bit-true model, we sweep the fractional wordlength and the number of CORDIC micro-rotation stages against all 11 matrices in `Final11Pattern.mat`. A configuration is accepted only when the RMSE of eigenvalues and eigenvectors both remain below 10^{-2} for every test pattern:

$$\mathrm{RMSE}_\lambda = \sqrt{\frac{1}{3}\sum_{i=1}^{3}(\hat{\lambda}_i - \lambda_i)^2} \quad \mathrm{RMSE}_v = \sqrt{\frac{1}{9}\sum_{i=1}^{3}\|\hat{\mathrm{v}}_i - \mathrm{v}_i\|_2^2}$$

where λ_i and $\hat{\lambda}_i$ are the reference and bit-true eigenvalues, and v_i and $\hat{\mathrm{v}}_i$ are the associated eigenvectors.
We use a two-step search rather than tuning both knobs at once. First, with the CORDIC stage count fixed at 8, we increase the fractional wordlength b until the RMSE criterion is met on all 11 sets; the smallest such value is $b = 11$. Second, with $b = 11$ held fixed, we sweep the CORDIC stage count and find that 8 stages is again the minimum setting that satisfies the same criterion. Thus, the selected bit-true configuration is **11 fractional bits** and **8 CORDIC stages**, which are adopted consistently in the subsequent verification and hardware implementation.

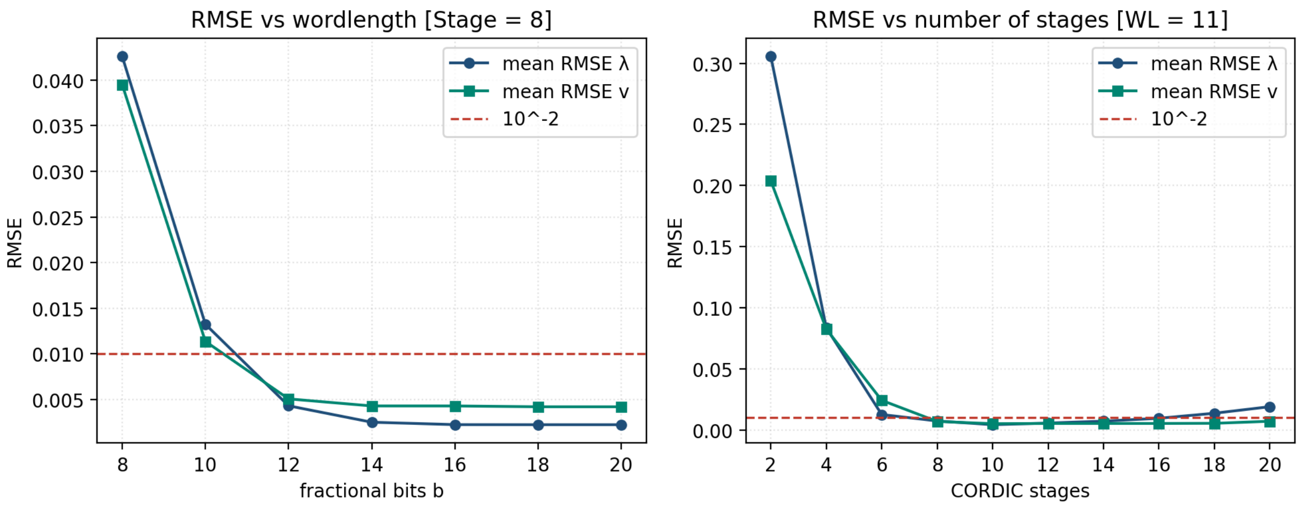

Figure 1.1: RMSE vs. fractional bits ($N_{\mathrm{stage}} = 8$, left) and CORDIC stages ($b = 11$, right).

The two-step sweeps above are useful for intuition, but they do not jointly identify the smallest (b, N_{stage}) pair. Each pass fixes one parameter at the value found earlier, so a configuration that fails in isolation may still satisfy the RMSE criterion when both knobs are set together, and the true minimum combination may be missed.

We therefore perform a two-dimensional sweep over fractional bits b and CORDIC stages N_{stage}. Every grid point is evaluated on all 11 test patterns; a setting passes only when both RMSE_{λ} and RMSE_{v} remain below 10^{-2} for every set.

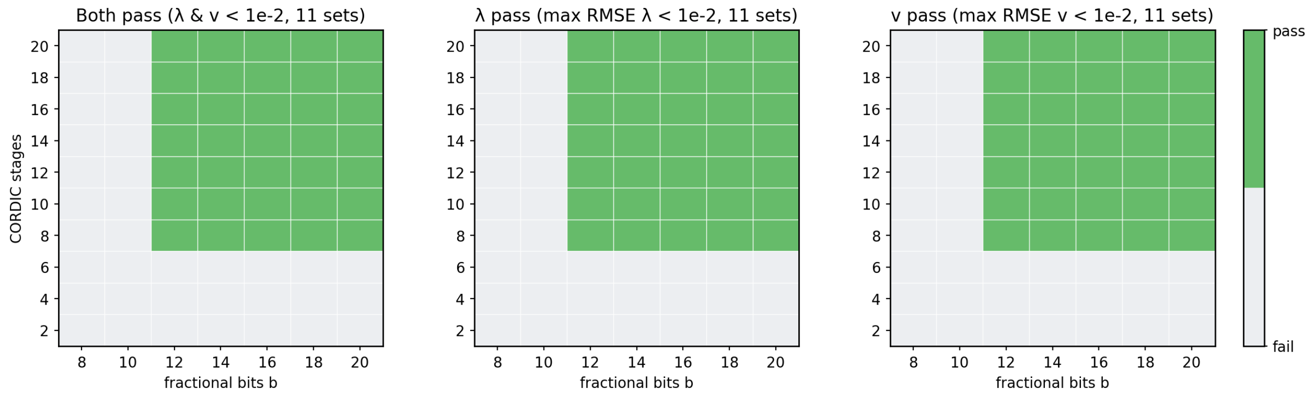

Figure 1.2: 2D pass maps over b and N_{stage} (11 test patterns).

Table 1.1 summarizes the hardware configuration selected from the sweeps above.

Table 1.1: Hardware Configuration

Parameter	**Value**
`B` (fractional bits)	11
`N_STAGES` (CORDIC stages)	8
`MAX_ITER` (max iterations)	7

1.2 Integer Bit-Width Analysis

With the configuration in Table 1.1, we record the dynamic range of each EVD stage across all 11 patterns and derive the minimum integer wordlength needed at each node. Table 1.2 lists the worst-case integer bit requirement and the corresponding fixed-point format (1 sign bit + integer + B fractional bits).

Table 1.2: Integer Bit-Width Analysis

Node	**Min**	**Max**	**Max abs**	**Int. bits**	**Q format**	**Total**
Quantized input	−2.394	10.239	10.239	4	Q4.11	16
Matrix T/R	−5.117	10.585	10.585	4	Q4.11	16
Eigenvector U	−0.981	1.000	1.000	1	Q1.11	13
CORDIC input	−5.117	10.585	10.585	4	Q4.11	16
CORDIC internal	−10.586	17.432	17.432	5	Q5.11	17
Scaled output	−5.117	10.585	10.585	4	Q4.11	16

1.3 Fixed-Point Bit-True Model

With the fractional wordlength, CORDIC stage count, iteration limit, and integer bit-width requirements established above, we implement a fixed-point bit-true model of the iterative QR EVD datapath. The model applies the formats in Tables 1.1 and 1.2 at each quantization point so that its arithmetic matches the behavior targeted by the RTL design.

This model supports more accurate fixed-point error analysis than a floating-point reference, because quantization, CORDIC rounding, and iterative accumulation are modeled explicitly. It also supplies golden eigenvalues and eigenvectors for RTL verification, so behavior and gate-level simulations can be compared against a consistent fixed-point reference instead of double-precision software alone.

1.4 CORDIC Scaling Factor

Sections 1.1–1.2 do not analyze the impact of quantizing the CORDIC scaling factor. They model gain compensation as $Q_B(X/K)$ only (divide by ideal K), then quantize the quotient. This is intentional, we do not want scaling-factor precision to enlarge global B and widen the entire systolic datapath. Therefore, Extra bits are spent only at this multiply-and-round point, not across the systolic array.

Table 1.3: CORDIC Scaling Factor Configuration

Parameter	**Value**
`DATA_FRAC` (datapath fractional bits)	11
`DATA_WIDTH` (datapath total bits)	17
`K_INV_FRAC` (fractional bits for $1/K$)	16
`K_INV_DW` (total bits for $1/K$)	17

Section 2: Bit-True Model Verification

We evaluate the bit-true model on all 11 patterns in `Final11Pattern.mat` with $B = 11$, $N_{\text{stage}} = 8$, and $\text{MAX\_ITER} = 7$. Figure 2.1 plots RMSE_λ and RMSE_ν per pattern; both remain below 10^{-2}.

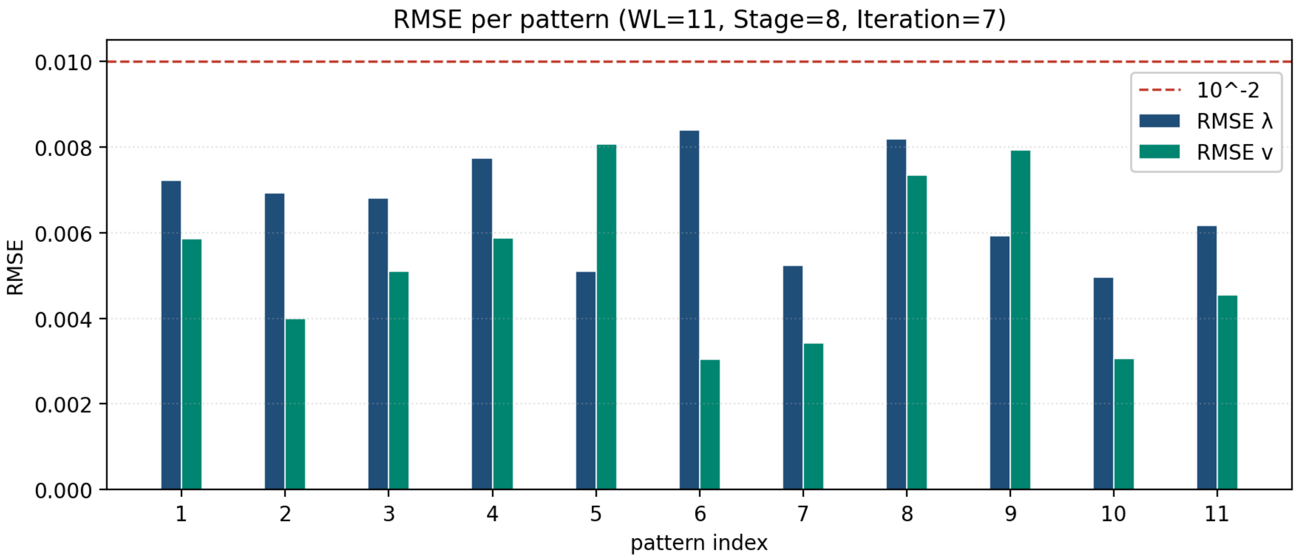

Figure 2.1: RMSE per pattern.

Section 3: Hardware Architecture and Control

Section 4: Hardware Simulation

Section 5: Fixed-Point vs. Hardware Consistency

Section 6: Area-Time Product

Section 7: Design Innovations and Team Contributions

7.1 Team Contributions

Table 7.1: Team Contributions and Work Distribution

Work Item	MING-HONG, LIN	YING-CHENG, CHEN
Algorithm Research and Floating-Point Modeling	✓	✓
Bit-True Model Development	✓	✓
Hardware Word-Length and Accuracy Analysis	✓	✓
Hardware Architecture and Control Flow Design	✓	✓
RTL Design	✓	✓
Testbench Development and Unit Verification	✓	✓
Behavioral Simulation	✓	✓
Logic Synthesis and Area/Timing Analysis	✓	✓
Post-Synthesis Simulation	✓	✓
Hardware Optimization	✓	✓
Area-Time Product Evaluation	✓	✓
Result Analysis and Visualization	✓	✓
Report Writing	✓	✓
Final File Organization and Submission	✓	✓